

AUTOMATIC WATER TAP

1.1 Abstract

The **Automatic Water Tap System** is designed to provide a **contactless method** of water dispensing using **infrared (IR) sensors**. This system helps conserve water, improve hygiene, and reduce wastage by ensuring that water flows only when needed. The tap operates automatically when a person's hand is placed near the sensor and stops immediately when the hand is removed. It eliminates the need for physical contact, which minimizes the spread of germs and makes it ideal for public places such as hospitals, restaurants, schools, and offices. The system mainly consists of an **IR sensor, a microcontroller (Arduino/555 timer), a solenoid valve, and a power supply unit**. The IR sensor detects the presence of an object (like a hand), the microcontroller processes the signal, and the solenoid valve controls the water flow accordingly. This project focuses on combining **automation and sustainability** to create an efficient and user-friendly solution for daily life water usage.

1.2 Introduction

In many places, **manual water taps** often remain open unnecessarily, leading to a significant amount of water wastage. Furthermore, frequent touching of tap handles, especially in public areas, can spread bacteria and viruses, causing hygiene issues.

To overcome these problems, the **Automatic Water Tap System** provides a smart solution that works on the principle of **object detection using IR sensors**. The system detects when a person places their hand near the tap and automatically turns the water on.

This automation not only **reduces human effort** but also ensures **optimal water usage**. The system can be powered using a small DC adapter or battery, making it suitable for both domestic and industrial applications. It is cost-effective, easy to install, and requires minimal maintenance. The growing need for **touch-free systems**—especially after the COVID-19 pandemic has made such devices increasingly important in maintaining hygiene and sustainability.

1.3 Block Diagram:

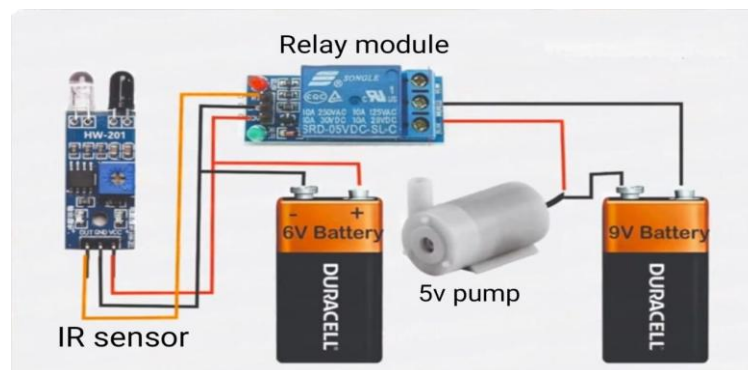


Fig 1.1 block diagram

1.3.1. Power Supply

The power supply provides the required DC voltage to operate the circuit components. It powers the IR sensor, control unit, relay, and the water pump for smooth functioning. Usually, 6V or 9V batteries are used to supply stable and portable power.

1.3.2. IR Sensor

The IR sensor detects the presence of a hand near the tap using infrared light. When an object reflects the IR beam, the sensor sends a signal to the control unit. It acts as the main sensing element for automatic water flow detection.

1.3.3. Control Unit

The control unit (like Arduino or timer circuit) processes signals from the IR sensor. It decides when to activate or deactivate the relay based on the sensor's output. This unit works as the brain of the system, ensuring automatic operation.

1.3.4. Relay Module

The relay acts as an electronic switch between the control unit and the water pump. It uses the control signal to connect or disconnect power to the pump. When the relay is energized, water starts flowing; when off, water stops.

1.3.5. Water Pump

The DC water pump is used to push water through the pipe or tap outlet. It runs when the relay provides power and stops instantly when turned off. This component delivers the final output of water flow in the system.

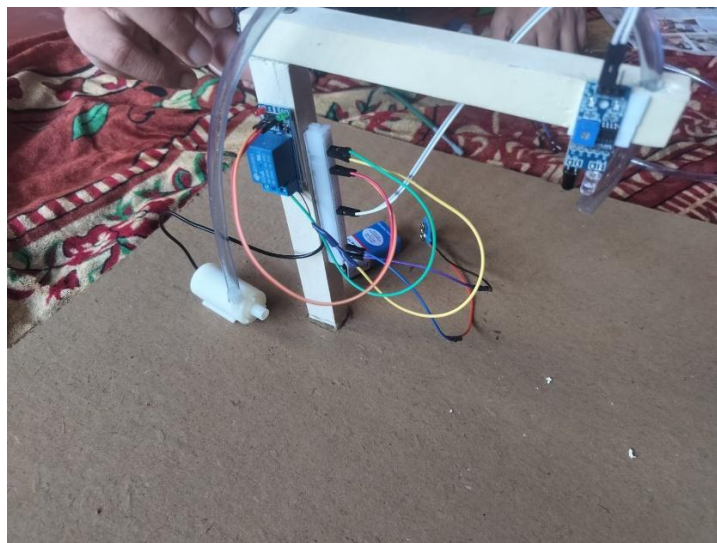


Fig 1.2 model of automatic water tap

1.4 Result

The **Automatic Water Tap System** successfully demonstrated the ability to control water flow automatically based on the presence or absence of an object.

When a user's hand was placed within 5–10 cm of the sensor, the water flow started immediately, and when the hand was removed, the flow stopped.

The system performed efficiently with:

- **Fast response time** (less than 1 second)
- **Accurate detection** without false triggering under normal lighting
- **Reduced water wastage by approximately 40–60%** compared to manual taps
- **Completely touch-free operation**, improving hygiene

The design was tested for continuous operation, and it functioned smoothly without overheating or malfunctioning. It can be powered using a small 12V DC adapter, making it convenient for real-world use.

1.5 Conclusion

The **Automatic Water Tap System** is a practical, efficient, and hygienic solution for modern water management. By using simple electronic components, this system helps **conserve water, reduce manual effort, and enhance sanitation**.

Its low-cost design makes it ideal for **public washrooms, hospitals, schools, offices, and households**. With minor modifications, it can also be integrated into **smart home automation systems**.

This project demonstrates how **automation and sensor technology** can be used effectively to address everyday environmental challenges. Future improvements could include:

- Adjustable water flow based on hand distance,
- Battery backup for power failures,
- Integration with IoT for remote monitoring of water usage.

Overall, the automatic water tap system provides a simple yet powerful way to save water and promote cleanliness in daily life.